

AICHU TAN
Data Scientist | AI Engineer

Dublin, CA | 408-650-2150 | thedebbietan@gmail.com | Portfolio | LinkedIn

PROFESSIONAL SUMMARY

Data Scientist & AI Engineer with an M.S. in Big Data Analytics (GPA 4.0) and 12+ years of prior experience spanning statistical analysis, process optimization, and business analytics. Proven track record of designing and deploying end-to-end ML and LLM systems, including a YOLOv8 computer vision and RAG-powered platform that improved plant disease detection performance from 70% to 89.4% mAP@50. Skilled in time-series forecasting (R² up to 0.89), complex SQL/Python ETL pipelines, and cloud-based MLOps deployment.

TECHNICAL SKILLS

Programming: Python, SQL(CTEs, Window Functions, Joins), R
AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, YOLOv8, LangChain (RAG), LLMs, Time Series (ARIMA/SARIMAX), K-Prototypes
Data Engineering & Cloud: AWS (EC2, S3, RDS), ETL Pipelines, Data Modeling, Supabase, Render, Git, MLflow
Analytics & Visualization: Pandas, NumPy, Feature Engineering, EDA, Streamlit, Tableau, ArcGIS

PROFESSIONAL EXPERIENCE

Data Scientist

Sep 2025 - Present

SDSU DiMolab : https://aichutan.github.io/Dengue_Taiwan_Forecast/

- Designed end-to-end SQL and Python ETL pipelines to ingest and standardize 27+ years of longitudinal meteorological, entomological, and population data for outbreak modeling.
- Developed automated data-quality validation rules to flag anomalies and filter reporting noise before downstream modeling.
- Diagnosed instability in daily-frequency case counts and re-aggregated data into 3-day and 5-day windows, stabilizing forecasts and achieving R² up to 0.89, enabling earlier, more targeted public health response.
- Deployed an interactive forecasting application using Streamlit, Supabase, and Render Cloud, enabling users to explore forecasts, compare model performance, and visualize prediction results.
- Co-authoring a research manuscript on outbreak forecasting for submission to a peer-reviewed journal.

AI Engineer

Feb 2025 - Dec 2025

SDSU Metabolism of Cities Living Lab : https://aichutan.github.io/ai_for_farmers/

- Designed and deployed an interactive AI platform on Hugging Face Spaces using YOLOv8, LangChain RAG, and Streamlit, enabling real-time plant disease diagnosis and AI-powered treatment recommendations.
- Analyzed model performance across 35,000 plant disease images using confusion matrix analysis and improved annotation quality for 13,500+ images, increasing detection performance from 70% to 89.4% mAP@50.
- Built an LLM-powered retrieval system grounded in agricultural research literature to minimize hallucinations and deliver reliable, evidence-based recommendations while reducing API costs.
- Co-authored research accepted at the 4th International One Health Conference (Rome, 2025).

Business Owner

Apr 2009 - Oct 2016

Wholesale Clothing

- Increased annual sales 50% through customer segmentation, market analysis, and demand forecasting.
- Optimized inventory planning and supply chain operations using sales forecasting and purchasing analytics.
- Managed international sourcing and import/export operations across Taiwan, China, and Indonesia.

Process Engineer

Jul 2004 - Feb 2009

ProMOS Technology

- Engineered statistical process control (SPC) and root-cause analysis frameworks on high-volume production lines to detect wafer variation, optimize yield, and stabilize DRAM manufacturing processes.
- Developed and optimized low-temperature barrier nitride processes, analyzing wafer-level experiment data to enhance transistor performance and process stability.
- Directed international technology transfer and tool qualifications, utilizing cross-functional statistical evaluations to ensure equipment reliability and seamless production ramp-up.

SELECTED DATA SCIENCE PROJECTS

Predicting Restaurant Ratings Using Yelp Metadata : https://aichutan.github.io/BDA602_final_yelp_analysis/

Python • SQL • XGBoost • Random Forest • MLflow

- Developed an end-to-end machine learning classification pipeline using 36,261 Yelp restaurant records to predict high- versus low-rated businesses based on operational attributes rather than review text.
- Engineered 61 predictive features from business metadata, including operating hours, service offerings, and accessibility; achieved 73.3% accuracy and 0.79 ROC-AUC using XGBoost, with experiment tracking in MLflow.
- Found via K-Prototypes clustering to identify customer behavior patterns, revealing that accessibility features were stronger predictors of restaurant ratings than price tier, providing actionable business insights.

Online Ticket Reservation System : https://aichutan.github.io/Online-Ticket-Database-Management-System/

MySQL • Python • AWS RDS

- Designed a normalized relational database (ERD → schema) with trigger-based validation to guarantee transactional data integrity across bookings and payments.
- Deployed the database on AWS RDS with role-based access control, enabling secure and scalable data management.
- Built a Python/Datapane dashboard surfacing \$70K+ in revenue trends and customer behavior patterns for business decision-making.

Mapping for Change - Urban Inequality Analysis : https://aichutan.github.io/Mapping-for-change/index.html

Spatial Analysis • Time Series • Data Visualization

- Integrated five public datasets (ACS, 311, PITC, and Tree Equity Score) into an interactive ArcGIS dashboard to analyze housing burden, homelessness, and environmental equity across urban neighborhoods.
- Performed exploratory data analysis and spatial statistical analysis to identify significant geographic clusters linking housing cost burden with reduced tree equity, supporting targeted policy recommendations.

RESEARCH & PUBLICATIONS

Tan A., Vito D., Fernandez G., 2025. "AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System" One Health International Conference, Università Campus Bio-Medico di Roma, September 30-October 2, 2025 Rome, (Italy).
D. Hamza, C. Wang, P. Lin, C.-K. Kao, C. Kuo, A. Ku, 2007. "Influence of Low Temperature Barrier Nitride Film on Transistor Characteristics for Advanced DRAM" 24th International VLSI Multilevel Interconnection Conference (VMIC 2007).

EDUCATION

M.S. in Big Data Analytics, San Diego State University - GPA 4.0
B.S. in Chemical Engineering, National Cheng Kung University (Taiwan)

Aug 2024 - Dec 2025
Sept 1999 - June 2003